



Guideline

Subcutaneous Administration of Antibiotics

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Hospital-Wide	Medical, Nursing, Pharmacy

Introduction

Subcutaneous administration of antibiotics serves as an alternative to intravenous antibiotic administration.

Some examples when this option may be the preferred route of administration include (but not exclusive to)

- Patient with difficulty in obtaining vascular access
- Patient where there is difficulty with maintaining an IVC in situ (relating to cognitive or behavioural disorders)
- In situations that may prevent unnecessary hospital admission or help facilitate hospital discharge

There is evidence demonstrating the safety and efficacy of this form of antibiotic delivery for ceftriaxone, teicoplanin and ertapenem.

This guideline will refer to those three antimicrobials; ceftriaxone, teicoplanin and ertapenem, as these agents have comparative studies demonstrating similar efficacy between the subcutaneous and intravenous routes.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☒

Definitions

Pharmacokinetics	A term that describes how the human body interacts with a drug after administration, with particular focus on four stages; absorption, distribution, metabolism and excretion.
Pharmacodynamics	The term used to describe the effect of the drug in the body. It studies the relationship between the concentration of the drug at the site of action and the biochemical and physiological effect.
Subcutaneous administration	The insertion of a medication beneath the skin, into the subcutis (layer of skin directly below the dermis and epidermis) using a hypodermic needle and syringe
SC	Subcutaneous
Isotonic	Term to describe a solution that has the same osmotic pressure or solute concentration as another solution – in the context of human biology, isotonicity would be in reference to equal osmotic pressure to typically body fluids and cells
Hypertonic	Term to describe a solution that has a higher osmotic pressure or solute concentration than another solution – in this context it would be body fluid or intracellular fluid

Principles of subcutaneous antibiotic administration

- The subcutaneous route of administration of antibiotics will be restricted to 3 antimicrobials at this stage based on the evidence currently available.
 - Ceftriaxone**
 - Teicoplanin**
 - Ertapenem**
- Consideration of the subcutaneous route for administration of one of the approved antibiotics will require discussion with an Infectious Diseases Physician.
A request to administer any other antimicrobial via subcutaneous route will require prior approval from the Medicines and Therapeutics Committee via an IPA submission on the WAIPAS platform [WAH IPA PROD \(hcswebsites.com.au\)](https://hcswebsites.com.au)
- The first dose should be administered whilst the patient is an inpatient to ensure that they are able to tolerate the dose

Important considerations

- i. Ensure that the patient is suitable for subcutaneous administration.
 - a. Is the intended site of administration (usually the lateral abdominal wall or thigh) suitable for repeated subcutaneous administration?
 - b. Are there any existing dermatological conditions present that may affect the skin integrity or predispose to a compromised skin barrier at the intended injection site?
- ii. In comparison to the intravenous route, the bioavailability of the drug via the subcutaneous route may take longer and be incomplete as the drug must cross the subcutaneous tissue before reaching the systemic circulation. The PK of individual drugs between IV and SC routes of administration can vary.

GUIDELINE

The process for subcutaneous administration of antibiotics below should be followed in conjunction with the SCGOPHCG Nursing Practice Guideline (Reference # NPG32) for Subcutaneous Infusions and Subcutaneous Fluid Administration

https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/NMAHS/SCGH/SCGOPHCG.NPG.Subcutaneous%20Infusions_and%20Subcutaneous_Fluid_Administration.pdf

Patient preparation

1. Select the site of subcutaneous administration for the patient.
Common sites include the abdomen (at or under the level of the belly button), the anterior or lateral aspects of the thighs, outer arms, scapula, or upper anterior chest wall (away from axilla) Appendix 2.
2. Assess the skin for the presence of infection, broken skin or evidence of previous irradiation treatment that would preclude use of the subcutaneous route.
3. The site selected should be easily accessible and have an area with good depth of subcutaneous fat and be distant from a joint or any bony prominence. It should also be easily accessible, with consideration of patient factors (e.g. patient ambulatory/mobility status, comfort level, degree of agitation or distress).
4. Assess the patient for generalised body oedema or poor peripheral circulation as these factors may impact on the subcutaneous delivery of the drug.
5. The approved subcutaneous catheters at SCGOPHCG (BD Saf-T-Intima™) can remain in place for up to 7 days if not causing patient irritation. If the catheter needs to be re-sited then the location should be rotated to a different area.
6. The site of subcutaneous administration must be reviewed on each occasion prior to drug/fluid administration to assess for any inflammation, pain or leakage.

Administration preparation

****Please refer to the SCGOPHCG nursing practice guideline (NPG32) alongside this guideline for administration details****

1. Daily assessment of insertion site is required.
2. Limited evidence suggests that injections into the thigh are more painful than the abdomen.
3. To improve tolerability the infusion site should be rotated every 72 to 96 hours and the use of non-rigid catheters (20G to 27G).
4. Rate of injection has not been shown to impact on injection-related pain.
5. Presence of antimicrobial preservatives in drug formulations may also contribute to local discomfort and pain at the injection site. If causing concern, seek a formulation absent of antimicrobial preservatives (if available).

In the event the patient is unable to tolerate the subcutaneous infusion due to infusion-related irritation or pain, minor amendments to the general guide may be made following discussion with the pharmacist to help reduce any patient discomfort and improve tolerability of the infusion.

Approved antimicrobials for subcutaneous administration

1. Ceftriaxone

Approved dose for subcutaneous administration

- 1g daily

Preparation advice

- Dissolve 1g of ceftriaxone in 10mL water for injections or sodium chloride 0.9%
- Shake well to dissolve
- Dilute the contents further to a final volume of 50mL using sodium chloride 0.9%*

**Glucose 5% may be used as a substitute for sodium chloride if preferred or clinically warranted.*

Administration

- To be administered as a subcutaneous infusion via subcutaneous butterfly catheter over 30 minutes.

Stability

- Reconstituted solution of ceftriaxone will remain stable for a period of 6 hours when stored below 25°C, or for 24 hours if stored at 2 to 8°C (refrigerated).

2. Ertapenem

Approved dose for subcutaneous administration

- 500mg to 1g daily

Preparation advice

- Dissolve 1g of ertapenem in 10mL water for injections or sodium chloride 0.9%
- Shake well to dissolve
- Dilute the contents further to a final volume of 50mL using sodium chloride 0.9%*

**Glucose 5% may be used as a substitute for sodium chloride if preferred or clinically warranted.*

Administration

- To be administered as a subcutaneous infusion via subcutaneous butterfly catheter over 30 minutes.

Stability

- Reconstituted solution of ertapenem will remain stable for a period of 6 hours when stored below 25°C, or for 24 hours if stored at 2 to 8°C (refrigerated).
- Use within 4 hours of removal from the fridge.

3. Teicoplanin

Approved dose for subcutaneous administration

- 400mg to 800mg daily (6 to 12mg/kg/24hr)

Preparation advice

- All teicoplanin vials are supplied with dedicated diluents.
- There is slight variance in the volume of diluent depending on the brand of teicoplanin.

Targocid® brand – 3.14mL of water for injections

Sandoz® brand - 3.2mL of water for injections

- Reconstitute each vial using the corresponding supplied diluent.
- The diluent should be added slowly down the wall of each vial.
- Do NOT shake the vials.
- Instead, roll each vial gently between your palms until the powder has completely dissolved.
- If shaken, the solution will start to foam. If foaming occurs, stop agitating the solution. Let the vial stand aside for 15 minutes to allow the foam to settle.
- The final solution should be clear and colourless to yellow.
- Once completely dissolved, draw up the required dose from the vial/s (depending on the dose required) and dilute further to a final volume of 50mL using sodium chloride 0.9%*.

**Glucose 5% may be used as a substitute for sodium chloride if preferred or clinically warranted.*

Administration

- To be administered as a subcutaneous infusion via subcutaneous butterfly catheter over 30 minutes.

Stability

- Reconstituted solution of teicoplanin will remain stable for a period of 24 hours when stored at 2 to 8°C (refrigerated).
- Otherwise, the reconstituted solution should be used immediately after preparation or if removed from the fridge.

Monitoring and evaluation

- The SCGOPHCG Infectious Diseases team will maintain a record of patients that have been prescribed antibiotics for administration by the subcutaneous route.
- Record keeping will include the following fields
 - UMRN
 - Antimicrobial, brand and batch number
 - Dose
 - Dilution
 - Infusion rate
 - Duration of treatment
- The Antimicrobial Stewardship Committee (AMSC) and Medicines and Therapeutics Committee (MTC) will monitor compliance to this guideline by
 - Monitoring of SAC 1 and SAC 2 clinical incidents investigations relating to this practice
 - Review of individual clinical outcome reports to help assess the effectiveness of this practice

Related Documents

- Subcutaneous Infusions (Syringe Drivers) and Subcutaneous Fluid Administration. Reference # NPG32. Nursing Practice Guideline. Sir Charles Gairdner Osborne Park Health Care Group.

Acknowledgements

Subcutaneous Administration of Antibiotics guideline. Fiona Stanley Fremantle Hospitals Group 2021.

References

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2. Marie Jumpertz, Romain Guilhaumou, Matthieu Million, Philippe Parola, Jean-Christophe Lagier, Philippe Brouqui, Nadim Cassir, Subcutaneously administered antibiotics: a review, *Journal of Antimicrobial Chemotherapy*, Volume 78, Issue 1, January 2023, Pages 1–7

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5. Claire Roubaud-Baudron, Emmanuel Forestier, Thibaut Fraisse, Jacques Gaillat, Benoit de Wazières, Leonardo Pagani, Isabelle Ingrand, Louis Bernard, Gaëtan Gavazzi, Marc Paccalin, Tolerance of subcutaneously administered antibiotics: a French national prospective study, *Age and Ageing*, Volume 46, Issue 1, January 2017, Pages 151–155

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Standards Applicable	NSQHS Standards (Version 2): 				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

APPENDICES

Appendix 1. Subcutaneous administration guide

Guide for the Subcutaneous Administration of Select Antimicrobials					
Antimicrobial	Route	Dose	Preparation	Infusion Rate	Comments
Ceftriaxone	Subcutaneous infusion	1g daily	Dilute 1g vial using 10mL of WFI or NS Shake well Dilute further to total volume of 50mL using either NS or glucose 5%	Infuse 1g in 50mL solution over 30 minutes	Stable for 6 hours after reconstitution when left at room temperature
Ertapenem	Subcutaneous infusion	500-1000mg daily	Dilute 1g vial using 10mL of WFI or NS Shake well Dilute further to total volume of 50mL using either NS or glucose 5%	Infuse final solution over 30 minutes	Stable for 6 hours after reconstitution when left at room temperature Use within 4 hours after removal from the fridge
Teicoplanin	Subcutaneous infusion	Commonly 400-800mg daily (weight-based dosing; 6 to 12mg per kg/24hrs)	Reconstitute vial/s of teicoplanin using the supplied diluent by slowly adding down the wall of each vial Do NOT shake vials Gently roll vial/s between your palms to help dissolve the contents Once completely dissolved, withdraw the prescribed dose from the vial/s and dilute further to a total volume of 50mL using either NS or glucose 5%.	Infuse final solution over 30 minutes	If the solution starts to foam, stop agitating the solution and let the vial stand for 15 minutes for the foam to settle. Use the reconstituted solution immediately after preparation. The reconstituted solution is stable for only 24hrs stored between 2 to 8°C

Appendix 2. Recommended body sites for insertion of subcutaneous catheter

